

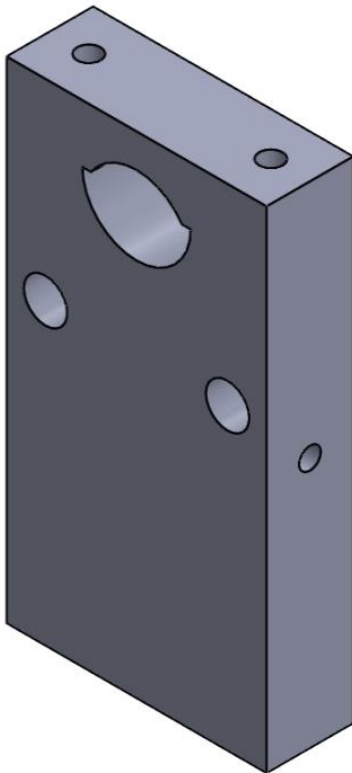
De – Buggas Fabrication Guide

P002 Leg Clamp Unit (Top and Bottom)

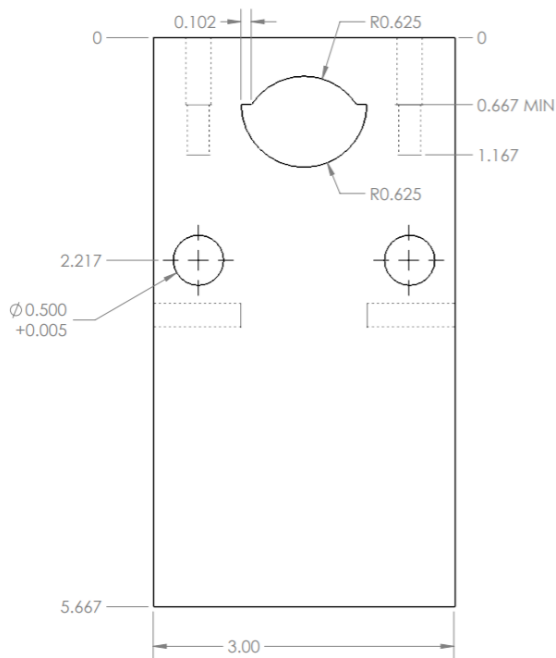
- 1) Start with 1" thick aluminum sheet metal



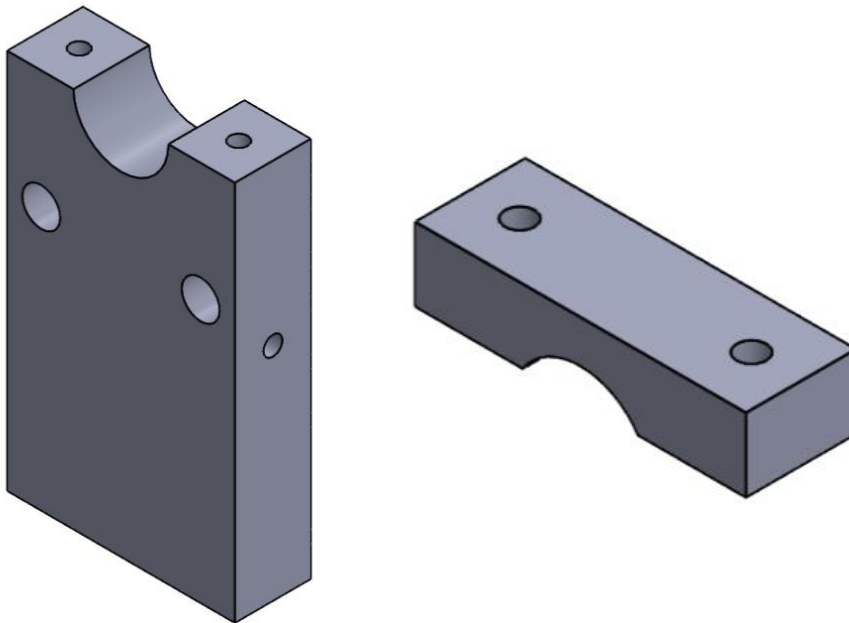
- 2) Waterjet a 5.667" x 3" rectangle from the stock



3) Waterjet both 0.5" holes and both adjacent semicircles



- 4) Tap 0.875" deep, 1/4-20 UNC holes centered on the left and right sides of the part.
- 5) On the top of the part, tap 1/4-20 UNC holes, 0.45 inches from either edge of the face, centered along the thickness of the plate.
- 6) On each of the holes created in 3. set a 0.266 inch counterbore no less than 0.667 inches deep from the face surface.
- 7) Waterjet a horizontal cut 0.500 inches from the top cutting P002 into P002Top and P002Bottom

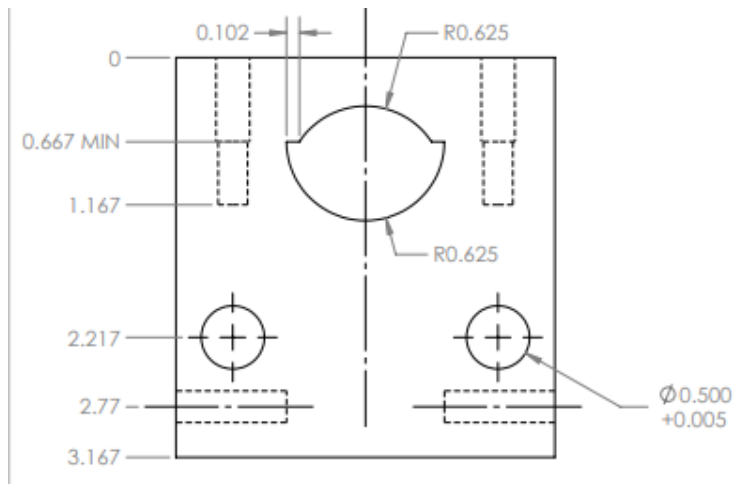


P002a Clamp Unit (Top and Bottom)

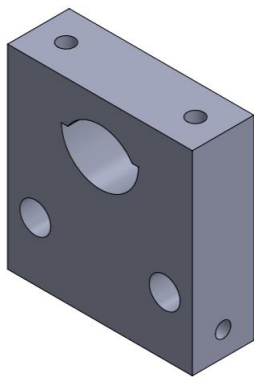
- 1) Start with 1" thick aluminum sheet metal



- 2) Waterjet a 3.167" x 3" rectangle from the stock

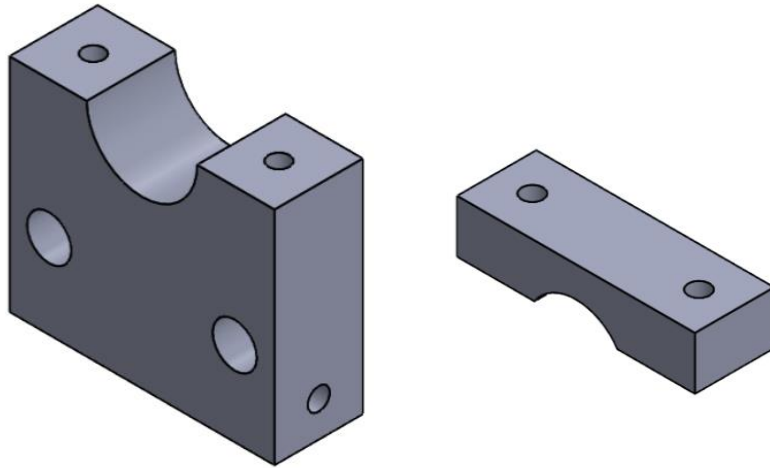


- 3) Waterjet both 0.5" holes and both adjacent semicircles.



- 4) Tap 0.875" deep, 1/4-20 UNC holes centered on the left and right sides of the part.
- 5) On the top of the part, tap 1/4-20 UNC holes, 0.45 inches from either edge of the face, centered along the thickness of the plate.

- 6) On each of the holes created in 3. set a 0.266-inch counterbore no less than 0.500 inches deep from the face surface.
- 7) Waterjet a horizontal cut 0.500 inches from the top cutting P002a into P002Top and P002aBottom

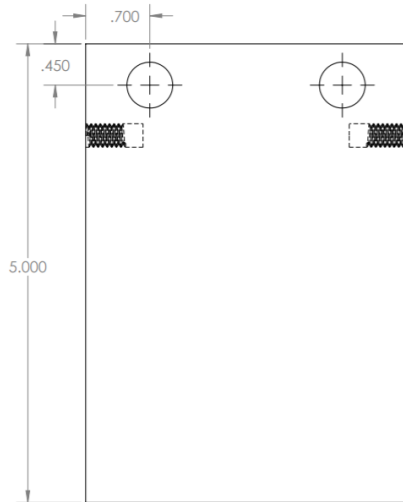


P004 Combined Leg Bracket

1. Start with 1" thick aluminum sheet metal



2. Waterjet a 3.5" x 5" rectangle from the stock
3. Waterjet both 0.5" holes .7" from the side and .45" from the top



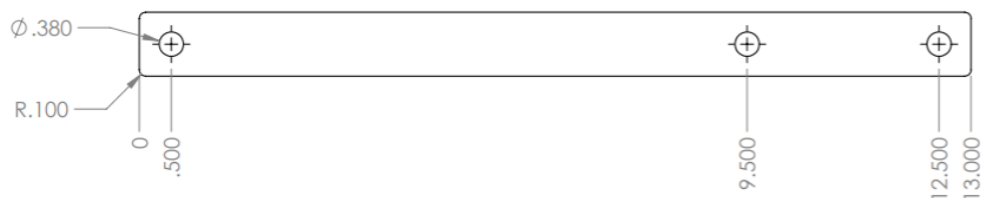
4. Drill and tap side holes $\frac{1}{4}$ - 20 UNC Threaded Blind to a $\frac{5}{8}$ " depth

P007 Lever Arm Long

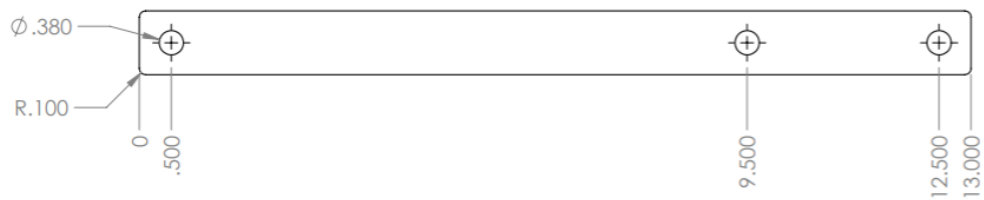
1. Start with 1, $\frac{1}{4}$ "x2"x24" piece of 6061 Aluminum Stock and cut to a length of 13.00"



2. Using a water jet, cut 2, 0.380" diameter holes, with the center of each hole positioned half and inch from each end of the stock and half an inch from either side so that it lies on the centerline



3. Using the same waterjet, cut another 0.380" diameter hole 9.5" from one end of the stock



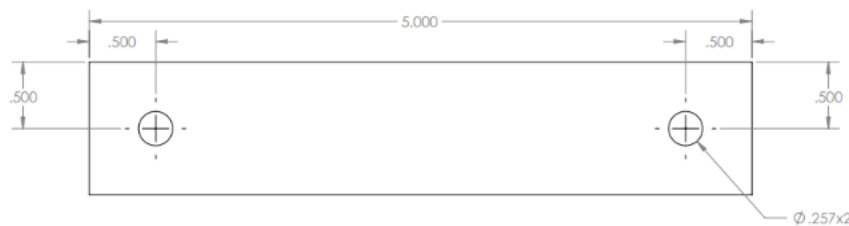
4. Using the same waterjet, cut a filet of radius 0.1" on each corner of the stock

P008 Short Lever Arm

1. Start with 1, 1/4"x2"x24" piece of 6061 Aluminum Stock and cut to length of 5.00"



2. Using a water jet, cut 2, 0.257" diameter holes, with the center of each hole positioned half an inch from each end of the stock and half an inch from either side so that it lies on the centerline

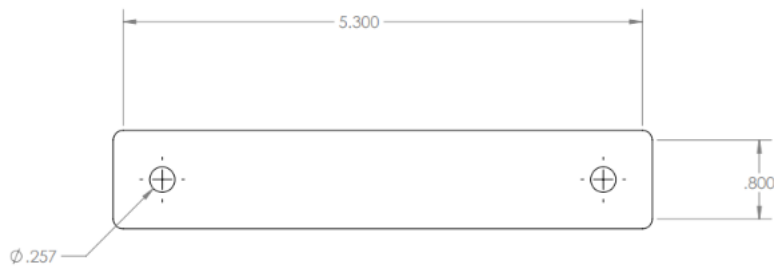


P010 Lever Clamp Spacer

1. Start with 1/4" Thick, 2"x24" and machine face from 1/4" to 1/8"



2. Place 1/8" sheet into water jet table
3. Use water jet to cut holes in the middle and fillet the edges

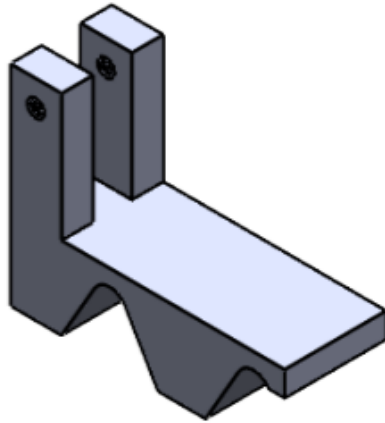


P011 Heating Jig

- 1) Start with 2x4x8 inch aluminum bar



- 2) Cut to Shape with water jet



- 3) Drill and tap $\frac{1}{4}$ -20 threaded holes centered 0.375" from the top of the arm through both sides

P013 Lever Handle

- 1) Start with 1" diameter wooden rod



- 2) Cut to length of 4"

